On 21 November 2016, the US Navy's US$ 4.4 billion guided missile destroyer USS Zumwalt, while traversing the Panama Canal, suffered a catastrophic propulsion failure;[1]just two days later, on 23 November, the British Royal Navy's US$ 1.2 billion hi-tech destroyer HMS Duncan suffered a catastrophic propulsion failure, while on NATO manoeuvres.[2]The reasons identified for this failure have been identified as what is called Chinese 'chip destroyers'—which are a type of microchip manufactured by the Chinese PLA, that the US Navy was forced to buy by the tens-of-thousands as a cost-cutting exercise.

The US Department of Defence's most critical electronics components are built to MilSpecs and are sourced from Taiwan, Japan and the US.[3] These microchips are supposedly built under rigorous military specifications to handle a nuclear environment. However, since the US lost its superconductor supremacy earlier in this century, globalisation has thoroughly muddied the marketplace. US and Japanese corporations like ADM, Motorola, Intel, NEC and Qualcomm have been outsourcing from their corporate subsidiaries and third-parties that supply low-cost chips from mainland China, with the parent company's brand on them.

When forensic technicians were deployed to back-engineer the faults on the USS *Zumwalt*, they found that all of the microchips received from a particular manufacturer had the same flaw.[4] Instead of being laser-etched with the MilSpec equivalent of an electromagnetic pulse (EMP), these had civilian wiring equivalent to a 3-amp fuse. In general, electronic components are connected together like a daisy-chain. Even if 10 hardened microchips soldered in a series survive a short burst of electromagnetic energy called an EMP—and this never happens—that component will fail. And if this happens during a war or conflict, there will be no replacements easily available. All the spares warehoused on a ship or ashore will also get roasted by the same invisible pulse or EMP. There are so many embedded semiconductors wired on a nanoscale that to rectify them, each one would have to be inspected and re-certified at test centres equipped with specialised equipment, which is an impossibility.[5]